

Hardeep Patel

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Summary

Founding engineer building **backend services**, **full-stack products**, and **production AI agents** on Azure and AWS. Works with enterprise customers from requirements through deployment, with depth in distributed systems and data pipelines.

Experience

Context Labs

Jun 2026 – Present

Founding Software Engineer, Forward Deployed

San Francisco, CA

- Cut customer time-to-value from **about 2 weeks to 2 days** by working directly with enterprise teams to define requirements, redesign onboarding, and carry solutions from architecture to production.
- Saved **20+ team hours/week across 5 production AI workflows** by designing agent memory that combines enterprise document retrieval (RAG), hot memory, and a knowledge graph, reducing retrieval-token usage.
- Engineered an **Azure email-ingestion pipeline** to turn years of customer email into context for AI agents, using Service Bus, Container Apps Jobs, PostgreSQL, and deduplicated Blob Storage.
- Standardized **OAuth 2.0 token management for 6+ integrations** by evaluating solutions and replacing custom code; made onboarding repeatable for Google Workspace, Microsoft 365, Slack, Salesforce, and Jira.
- Shipped the **customer-facing React/TypeScript platform and REST APIs**, plus the [company website](#).
- **Doubled response rates** by automating outbound prospecting to support customer acquisition.

Toya Agrisolutions

Dec 2023 – May 2024

Software Engineering Intern

Gujarat, India

- Reduced **API latency by 40%** and automated equipment booking for a platform with **500K+ registered farmers** by building Python and Node.js REST microservices on AWS Lambda backed by MongoDB.
- Forecasted demand with **LSTM models ($R^2 = 0.96$)**, **reducing stockouts by 15%**; integrated predictions into a React/D3.js dashboard deployed on AWS EC2 with Docker and CI/CD.

Infibeam Avenues

Jun 2023 – Aug 2023

Product Software Engineering Intern

Gujarat, India

- Automated payment-log monitoring across **2M+ transactions/month** using Node.js, AWS Lambda, S3, and CloudWatch; integrated Slack alerts that cut **incident response time by 27%**.
- Expanded **access-control compliance coverage by 40%** by implementing RBAC for Java/Spring Boot payment APIs with OAuth 2.0 and JWT-based single sign-on, in collaboration with product and cross-functional teams.

Indian Space Research Organization (ISRO)

Jan 2023 – Jun 2023

ML Engineer Intern

Gujarat, India

- Developed an **LSTM flood-forecasting pipeline** using 8+ environmental variables, achieving **88% accuracy** and **2-hour earlier warnings**; deployed model inference on AWS SageMaker through Flask and CI/CD.

Indian Institute of Technology (IIT), Roorkee

Jun 2022 – Jul 2022

Research Intern

Uttarakhand, India

- Optimized **C++ graph algorithms to run about 3x faster** by updating only affected subgraphs; preserved identical superbubble-detection results for dynamic graph analysis in pangenomics.

Projects

Distributed Banking System | Go, gRPC, Paxos/Multi-Paxos, Two-Phase Commit | [GitHub](#)

- Implemented a **fault-tolerant sharded banking system in Go** using Paxos/Multi-Paxos for consensus, gRPC for inter-shard communication, and two-phase commit for atomic transactions; benchmarked **80 TPS with linearizability**.

Autonomous Caching Policy Evolution | Python, C++, Rust, LLMs, RAG, ChampSim

- Improved **last-level cache hit rate by 11.09% over LRU** by building an LLM loop that generated and refined C++/Rust cache-replacement policies, using SQLite-backed retrieval and ChampSim simulations on HPC.

Technical Skills

Languages: Python, TypeScript, JavaScript, Go (Golang), C++, C, Java, SQL, HTML, CSS.

Backend & Web: FastAPI, Node.js, Express.js, Flask, Django, Spring Boot, REST APIs, gRPC, React, Next.js.

Data & AI: PostgreSQL, MongoDB, MySQL, pgvector, Kafka, Spark, RAG, LLM agents, LangChain, LangGraph, MCP.

Cloud & Delivery: Azure, AWS, GCP, Docker, Kubernetes, Terraform, CI/CD, Git, Linux, OAuth 2.0, JWT, SSO, RBAC.

Education

Arizona State University, Tempe, AZ

Aug 2024 – May 2026

M.S. in Computer Science | **GPA: 3.93/4.00** | Teaching Assistant: Cloud Computing and Math.

Relevant Coursework: Analysis of Algorithms, Distributed Systems, Statistical Machine Learning.

Pandit Deendayal Energy University, Gujarat, India

B.Tech. in Computer Science | **GPA: 9.61/10.00**

Jun 2020 – Jun 2024